Your Dreams Our Goal

POORNIMA UNIVERSITY

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Course Name: Basic of Mechanical Engineering

Topic: Aerodynamics

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2. Need for study of aerodynamics
3. Internal & External aerodynamics
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Objectives

1. Understand the need for studying aerodynamics, focusing on its significance in optimizing the performance and efficiency of vehicles, aircraft, and other structures.
2. Differentiate between internal and external aerodynamics, exploring their principles and applications in various industries and engineering designs.
3. Examine the basic geometry of an aerofoil and its role in influencing lift and drag forces, crucial for advancements in aviation and other aerodynamic systems.

Course Outcomes

CO-4: Illustrate the concepts of aerodynamics, electric vehicle architecture, and evaluate the performance of electric vehicles.

INTRODUCTION

- Aerodynamics is a sub-field of fluid dynamics and gas dynamics, and many aspects of aerodynamics theory are common to these fields.
- The term aerodynamics is often used synonymously with gas dynamics, the difference being that "gas dynamics" applies to the study of the motion of all gases, and is not limited to air.
- Aerodynamics is primarily concerned with the forces of drag and lift, which are caused by air passing over and around solid bodies.

INTRODUCTION

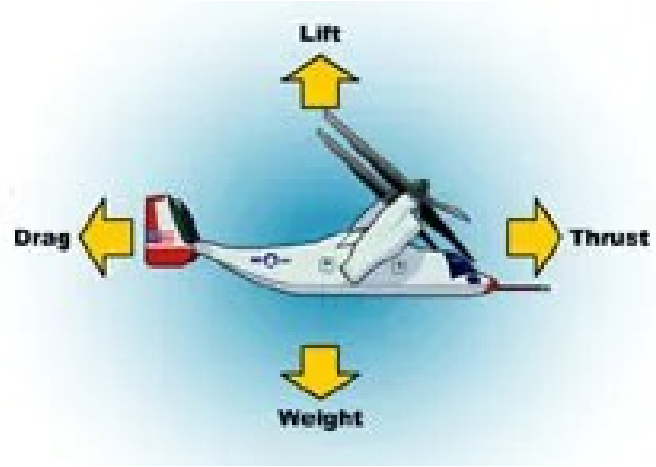
- Engineers apply the principles of aerodynamics to the designs of many different things, including buildings, bridges and even soccer balls; however, of primary concern is the aerodynamics of aircraft and automobiles.
- Aerodynamics comes into play in the study of flight and the science of building and operating an aircraft, which is called Aeronautics.
- Aeronautical engineers use the fundamentals of aerodynamics to design aircraft that fly through the Earth's atmosphere.

NEED FOR STUDY OF AERODYNAMICS

- Aerodynamics is the way objects move through air. The rules of aerodynamics explain how an airplane is able to fly.
- Anything that moves through air is affected by aerodynamics, from a rocket blasting off, to a kite flying. Since they are surrounded by air, even cars are affected by aerodynamics.
- Studying the motion of air around an object allows us to measure the forces of lift, which allows an aircraft to overcome gravity, and drag, which is the resistance an aircraft “feels” as it moves through the air.
- Everything moving through the air (including airplanes, rockets, and birds) is affected by aerodynamics.

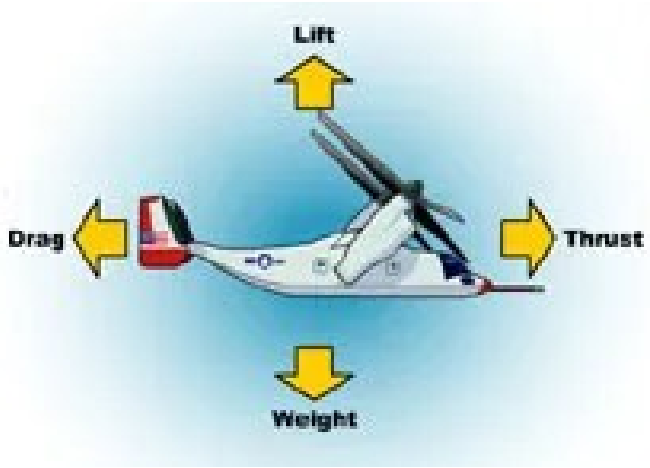
NEED FOR STUDY OF AERODYNAMICS

- ***What Are the Four Forces of Flight?***
  - The four forces of flight are lift, weight, thrust and drag. These forces make an object move up and down, and faster or slower.
  - The amount of each force compared to its opposing force determines how an object moves through the air.
- ***What is Weight?***
  - Gravity is a force that pulls everything down to Earth.
  - Weight is the amount of gravity multiplied by the mass of an object.
  - Weight is also the downward force that an aircraft must overcome to fly.
  - A kite has less mass and therefore less weight to overcome than a jumbo jet, but they both need the same thing in order to fly i.e. lift.



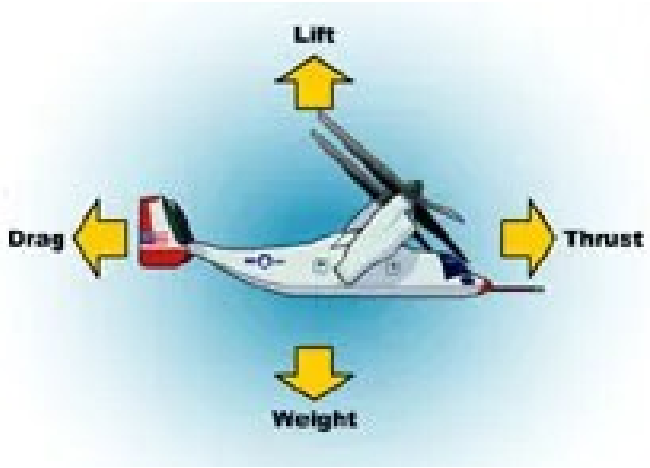
NEED FOR STUDY OF AERODYNAMICS

- ***What is Lift?***
  - Lift is the push that lets something move up. It is the force that is the opposite of weight.
  - Everything that flies must have lift. For an aircraft to move upward, it must have more lift than weight.
  - A hot air balloon has lift because the hot air inside is lighter than the air around it. Hot air rises and carries the balloon with it.
  - A helicopter's lift comes from the rotor blades. Their motion through the air moves the helicopter upward.
  - Lift for an airplane comes from its wings.



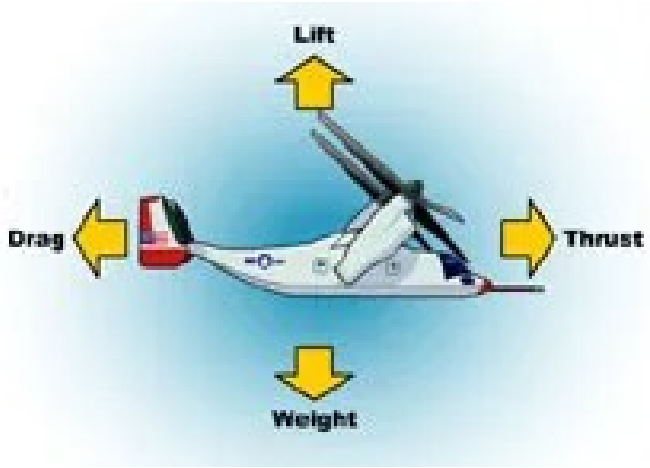
NEED FOR STUDY OF AERODYNAMICS

- ***How do an Airplane wings provide Lift?***
  - The shape of an airplane wings is what makes it possible for the airplane to fly.
  - Airplane wings are curved on top and flatter on the bottom. That shape makes air flow over the top faster than under the bottom.
  - As a result, less air pressure is on top of the wing. This lower pressure makes the wing, and the airplane it's attached to, move up.
  - Using curves to affect air pressure is a trick used on many aircraft.
  - Helicopter rotor blades use this curved shape.
  - Lift for kites also comes from a curved shape.
  - Even sailboats use this curved shape. A boat's sail is like a wing, that's what makes the sailboat move.



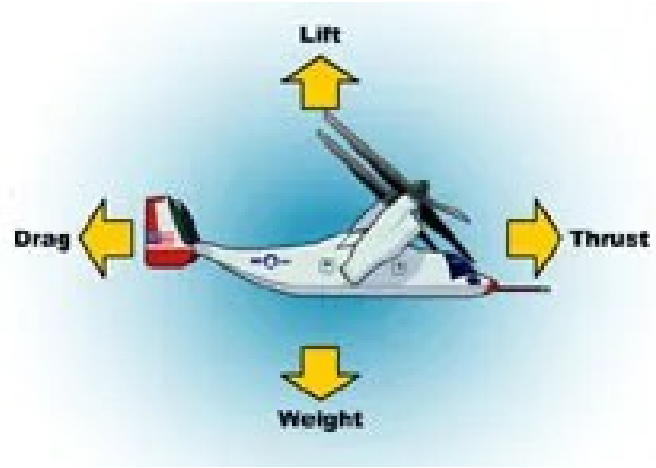
NEED FOR STUDY OF AERODYNAMICS

- ***What is Drag?***
  - Drag is a force that pulls back on something trying to move, it provides resistance, making it hard to move.
  - For example, it is more difficult to walk or run through water than through air.
  - Water causes more drag than air. The shape of an object also affects the amount of drag.
  - Round surfaces usually have less drag than flat ones.
  - Narrow surfaces usually have less drag than wide ones.
  - The more air that hits a surface, the more the dragthe air produces.



NEED FOR STUDY OF AERODYNAMICS

- ***What is Thrust?***
  - Thrust is the force that is the opposite of drag. It is the push that moves something forward.
  - For an aircraft to keep moving forward, it must have more thrust than drag.
  - A small airplane might get its thrust from a propeller.
  - A larger airplane might get its thrust from jet engines.
  - A glider does not have thrust. It can only fly until the drag causes it to slow down and land.



INTERNAL & EXTERNAL AERODYN..

- Aerodynamics is an applied science with many practical applications in engineering.
- No matter how elegant an aerodynamic theory may be, or how mathematically complex a numerical solution may be, or how sophisticated an aerodynamic experiment may be, all such efforts are usually aimed at one or more of the following practical objectives:

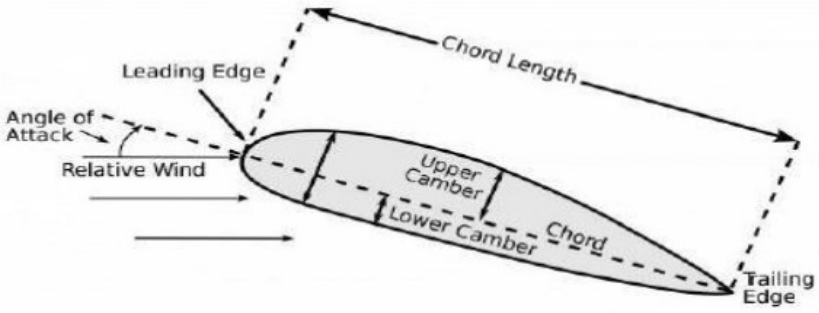
1) ***External Aerodynamics***:
   - The prediction of forces and moments on, and heat transfer to, bodies moving through a fluid (usually air).
2) ***Internal Aerodynamics***:
   - Determination of flows moving internally through ducts.

AERODYNAMICS IN DIFFERENT FIELDS

- Aerodynamics is important in a number of applications other than aerospace engineering.
  - It is a significant factor in any type of vehicle design, including automobiles.
  - It is important in the prediction of forces and moments acting on sailing vessels.
  - It is used in the design of mechanical components such as hard drive heads.
  - Structural engineers also use aerodynamics, and particularly aero-elasticity, to calculate wind loads in the design of large buildings and bridges.
  - Urban aerodynamics seeks to help town planners and designers improve comfort in outdoor spaces, create urban microclimates and reduce the effects of urban pollution.
  - The field of environmental aerodynamics describes the ways atmospheric circulation and flight mechanics affect ecosystems.
  - The aerodynamics of internal passages is important in heating/ventilation, gas piping, and in automotive engines where detailed flow patterns strongly affect the performance of the engine.

BASIC GEOMETRY OF AEROFOIL

- The leading edge is the point at the front of the aerofoil that has maximum curvature.
- The trailing edge is defined similarly as the point of maximum curvature at the rear of the aerofoil.
- The chord line is a straight line connecting the leading and trailing edges of the aerofoil.
- The chord length, or simply chord is the length of the chord line and is the characteristic dimension of the aerofoil section.



Conclusion

Here are some important pointers summarizing the key takeaways:

1. Understanding aerodynamics is essential for improving the performance, efficiency, and stability of vehicles, aircraft, and other structures interacting with air.
2. Internal aerodynamics deals with airflows inside objects like ducts or engines, while external aerodynamics focuses on the flow of air around objects like cars or planes.
3. Aerodynamics in different fields plays a crucial role in industries such as aviation, automotive, sports equipment design, and wind energy optimization.
4. The aerofoil shape influences lift and drag forces, making it a key component in aircraft wings and turbine blades.
5. Knowledge of aerodynamics is applied to enhance energy efficiency, reduce emissions, and improve safety in various engineering and environmental contexts.

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THANK YOU!!